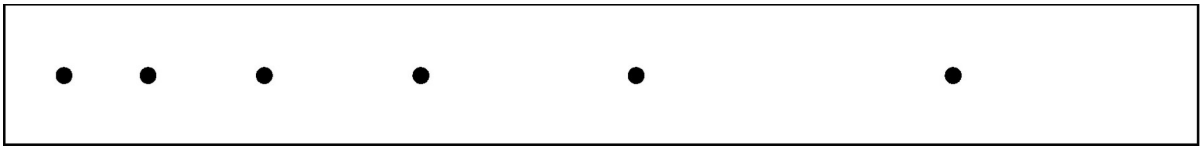
Ticker tapes

Answers

Describing motion

1. For each ticker tape shown, describe the motion of the body to which the tape was attached. The left-hand dot is at time zero.

(a)



The speed of the body is increasing with time. It is accelerating.

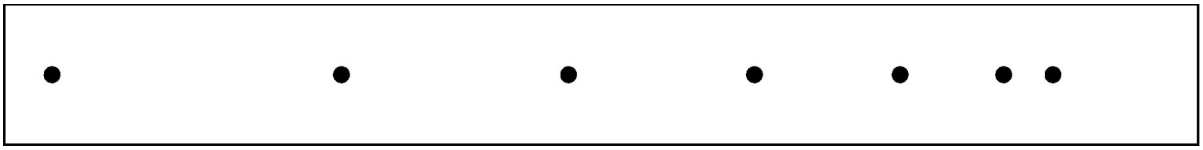
(b)



Initially, the body accelerates (its speed is increasing) and then rapidly decelerates (its speed is decreasing).

Measuring speed

2. The time between each of the dots on the following ticker tape is 0.02 s.



(a) Calculate the average speed of the object to which the tape was attached for each time interval. Express your answers in mm/s.

This is a sample calculation; student answers may vary depending on the magnification of the image.

Interval	Distance between dots (mm)	Average speed (mm/s)
AB	38	$\frac{38}{0.02}$ = 1900 mm/s
BC	30	$\frac{30}{0.02}$ = 1500 mm/s
CD	25	$\frac{25}{0.02}$ = 1250 mm/s
DE	20	$\frac{20}{0.02}$ = 1000 mm/s
EF	14	$\frac{14}{0.02}$ = 700 mm/s
FG	7	$\frac{7}{0.02}$ = 350 mm/s

(b) Describe the motion of the object to which this tape was attached.

The average speed of the object is decreasing with time. The object is decelerating.

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